PROJECT BRIEF · ANONYMIZED

Improving ASHA Member Survey Response Rates

Evaluating Survey Methodology Experiments

October 15, 2026 · @Rob

Project Background

The client is the American Speech-Language-Hearing Association (ASHA), a national professional membership association in the United States. Response rates to its member surveys have declined steadily over the past decade. The client wanted to identify effective practices from its own past survey methodology experiments to inform the design of its next survey round.

Note: Data come from experiment records provided by the client. Names of client staff have been omitted from this report.

Project Information	Details
Project type	Graduate student client consulting project, faculty-advised
Timeline	January – April 2026
Team	4 members
Client's core question	Which survey design choices increase member participation, and how should the next survey round be changed?
Tools	R; logistic regression, bootstrap, difference-in-differences, chi-square tests

Research Questions and Analytical Framework

We first screened the client's 30+ experiments from the past 20 years to identify the most impactful directions, then focused our in-depth analysis on two main workstreams.

1. **Review and screening.** We grouped the historical experiments into five categories: survey mode, reminders, incentives, visual design, and wording. Comparison showed that visual design and wording experiments mostly produced small between-group differences, survey mode showed the largest differences, and reminder strategy was the client's most pressing current cost concern.
2. **Workstream 1: Reminder strategy.** What are the response rates with 0, 1, and 2 reminder postcards? Does a second reminder add any further effect? Is there a short-term uptick in responses in the week after a reminder?
3. **Workstream 2: Survey mode.** How large is the response-rate gap between mail and web surveys? Do different member subgroups (employment status, facility type, region, professional role) differ in their sensitivity to mode? Does the responding sample still represent the full membership?

Key methodological judgment

Reminder postcards were sent only to people who had not yet responded, and these people are inherently harder to reach. Directly comparing those who did and did not receive a postcard would mistake *differences between the groups themselves* for *the effect of the reminder*. We leveraged the random assignment in the experiment to estimate the overall response rate under three scenarios—everyone receives no reminder, 1 reminder, or 2 reminders—and used a stratified bootstrap to construct confidence intervals. For the mode analysis, we used logistic regression with interaction terms plus likelihood ratio tests to determine whether the mode effect varies across subgroups.

Core Formulas

Counterfactual estimation of the reminder effect. Taking the response rate of the no-reminder group as the baseline, we add the increment contributed by each reminder wave:

$$RR_{\text{one}} = RR_B + (RR_1 - RR_0)$$

$$RR_{\text{two}} = RR_{\text{one}} + (RR_2 - RR_{1L})$$

RR_B = response rate of the no-reminder group; RR_1 = response rate among recipients of the first postcard; RR_0 = subsequent response rate among control-group nonrespondents over the same period; RR_2 = response rate among recipients of the second postcard; RR_{1L} = response rate among those who received only one postcard and were still nonrespondents at the second wave. Inference uses a stratified bootstrap (2,000 resamples).

Difference-in-differences for short-term effects. We compare average daily responses in the 7 days before and after each reminder:

$$\text{DiD} = (\bar{Y}^T_{\text{post}} - \bar{Y}^T_{\text{pre}}) - (\bar{Y}^C_{\text{post}} - \bar{Y}^C_{\text{pre}})$$

Mode × subgroup interaction model. A significant interaction coefficient indicates that the mode effect differs across subgroups; a likelihood ratio test (LRT) assesses whether adding the interaction term improves model fit:

$$\log \frac{p}{1-p} = \beta_0 + \beta_1 \,\text{Mode} + \beta_2 \,\text{Subgroup} + \beta_3 \,(\text{Mode} \times \text{Subgroup})$$

Analysis and Visualization Methods

Method	Purpose
Counterfactual estimation + stratified bootstrap	Estimate overall response rates and confidence intervals under 0 / 1 / 2 reminders
Chi-square test	Test differences in response rates between groups
Difference-in-differences (DiD)	Evaluate the short-term effect in the week after each reminder
Logistic regression with interactions + LRT	Determine whether the mode effect varies across subgroups
Grouped response-rate bar charts	Compare response rates across groups and modes
Sample composition comparison charts	Compare the responding sample with the target sample to assess representativeness
Choropleth map	Show response differences across the four U.S. Census regions

Key Findings

> **Bottom line:** The first reminder works; the second adds essentially nothing. Mail consistently outperforms web surveys on response rate, and matters especially for certain subgroups.

Reminder Strategy (2025 member survey, web questionnaire, n = 9,947)

+5.2 pp	+0.2 pp	+1.29 / day
First postcard 95% CI [3.6, 6.7], significant	Second postcard 95% CI [−1.1, 1.6], not significant	DiD: extra daily responses vs. control, week after 1st postcard

- **Direct group comparison:** postcard group 9.12% vs. control group 7.71%, a difference of +1.41 percentage points ($\chi^2 = 5.78$, $p = .016$).
- **Short-term effect (DiD):** in the week after the first postcard, average daily responses were 1.29 higher than in the control group; after the second postcard, 0.71 higher.

Scenario	Estimated overall response rate	95% confidence interval
No reminders	18.9%	[17.9%, 20.0%]
Up to 1 reminder postcard	24.1%	[22.6%, 25.7%]
Up to 2 reminder postcards	24.4%	[22.8%, 26.0%]

A point worth noting: Looking only at the raw response rate among postcard recipients, the reminder effect is just +1.41 percentage points; translated to the overall response rate using the counterfactual framework, the effect is +5.2 percentage points. The two calculations answer different questions, and we explained each separately to the client in our report.

Survey Mode

28.6%	13.5%	OR = 2.66
Mail response rate	Web response rate	Mail advantage, controlling for subgroups 95% CI [2.43, 2.91], $p < .001$

Subgroup dimension	Mail advantage over web	Interaction test	Conclusion
Employment status	Full-time +13.1 pp; part-time +20.6 pp	OR = 1.55 [1.27, 1.89], $p < .001$	Part-time members rely more on mail
Facility type	+11.7 pp (skilled nursing) to +19.2 pp (home health)	LRT $p = .024$; home health OR = 1.44, $p = .017$	Home health practitioners rely more on mail
Region (4 Census regions)	+14.4 pp to +15.6 pp	LRT $p = .762$	No difference
Professional role	Consistent across roles	LRT $p = .110$	No difference

Sample Representativeness

Compared with the target sample, the responding sample under-represents the Northeast by 2.26 percentage points and over-represents the Midwest by 1.58 percentage points; deviations for the South and West are under 1 percentage point. Differences in age composition are under 1 percentage point, and facility type and employment status are largely consistent. Observable bias is limited.

Recommendations to the Client

The core recommendation: **send one fewer reminder, keep the mail channel, and target sensitive subgroups.**

1. **Send only one reminder.** The first reminder already delivers most of the gain; the mailing cost of a second reminder buys almost no additional responses and can be cut.
2. **Do not switch entirely to web-only surveys.** Mail's response-rate advantage is stable and substantial; moving to web-only would lower the overall response rate.
3. **Allocate channels by subgroup.** Prioritize mail for part-time members and home health practitioners; otherwise these groups will be under-represented in the results. Region and professional role do not require differentiated treatment.
4. **Weigh the costs.** The reminder effect is real but modest in absolute terms; whether to deploy it at scale should be weighed against mailing costs.

My Contributions

My main responsibilities on this four-person project:

- **Client communication and problem definition:** worked with the client's research team to review 30+ historical experiments and narrow the focus to two workstreams: reminders and survey mode.
- **Reminder strategy analysis:** identified selection bias in the direct comparison and evaluated reminder effects using a counterfactual framework, stratified bootstrap, and difference-in-differences.
- **Mode analysis:** compared mail and web channels using logistic regression with interaction terms, broke results down by subgroup, and assessed sample representativeness.
- **Translating findings:** turned statistical results into reminder and channel recommendations the client could act on directly.

This project gave me practice with a complete analytical workflow: starting from a client's loosely defined problem, framing answerable research questions, choosing the right methods to avoid misleading conclusions, and arriving at actionable recommendations.